

Michael Bailey

AI Engineer — LLM Applications, Python & TypeScript

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PROFESSIONAL SUMMARY

Software engineer with 3 years at Sky, now leading AI adoption in the Global Reliability team. Built solo an AI-powered self-serve dashboard that replaces a BA-mediated change-request workflow consuming ~95 engineer-days/year: users describe a chart in natural language and drop returned widgets into their own dashboards, backed by a custom Model Context Protocol (MCP) server and agent orchestrator over OpenAI GPT. Previously owned Sky's internal monitoring dashboards — Next.js, GraphQL, OpenSearch — used across multiple departments. Software Developer Level 4 apprenticeship, Distinction.

KEY SKILLS

AI & LLM Applications: LLM application development, OpenAI (GPT) models, agentic workflow design, retrieval-augmented generation (RAG), LangGraph, Model Context Protocol (MCP), custom agent orchestration, tool use / function calling, prompt engineering, streaming responses (SSE), embeddings, vector search (OpenSearch)

Languages: Python, TypeScript (confident), JavaScript · Java, Golang, C# (familiar)

Frameworks & Platforms: Next.js, React, Node.js, FastAPI, Express, GraphQL, Recharts

Data & Search: OpenSearch (search, analytics, vector), PostgreSQL, Redis, MySQL, SQL

Cloud: Google Cloud Platform (GCP)

CI & Tooling: Jenkins (CI/CD), Docker, pytest, Jest; deployed on Sky's internal FluidCloud platform

Ways of Working: Technical leadership of AI initiatives · running internal AI onboarding sessions · production incident response (Global Reliability) · stakeholder demos across Sky teams and to Comcast executives

EXPERIENCE

Software Engineer — Sky UK

Sep 2022 – Present · London · Global Reliability

AI initiatives led:

- Designed and built solo an AI-powered self-serve monitoring dashboard: users ask questions like *"show me 4G usage split by tech in a stacked bar chart over the last eight months"* and a custom agent returns chart widgets they can resize, edit in-place, and assemble into shareable, per-user dashboards — collapsing a BA-mediated change-request cycle of 1-2 days per chart into sub-minute self-serve.
- Built a custom agent orchestrator (Python, uvicorn) running the multi-step tool-use loop client-side — Sky's internal AI gateway exposed only OpenAI Chat Completions, not the Responses API — paired with a custom MCP server (TypeScript) exposing OpenSearch as tools to it. Wired into a Next.js / React / TypeScript frontend; Dockerised and deployed via Jenkins on Sky's FluidCloud platform.
- Authored the compliance sign-off business case — covering security posture (prompt-injection boundaries, handle-based result retrieval to prevent raw-row leakage, tenant isolation), efficiency modelling (~35% projected gain against the ~95-day baseline), cost projection (~£100/month run-cost), and phased rollout.
- Ran 3 stakeholder demos during pre-rollout hardening: Sky's data engineering team, another Sky observability team, and a group of Comcast executives (Sky's parent company).
- Leading a department-wide shift to agentic workflows at my director's request — running onboarding sessions on using AI effectively and safely across engineering and non-engineering work.

Core engineering:

- Shipped an earlier reliability-insights tool for Sky's on-call rota — embedding-based retrieval over OpenSearch surfaces similar past incidents during triage, shortening time-to-context on repeat-class failures.

- Owned a suite of full-stack internal monitoring dashboards used across multiple Sky departments for 3 years (Next.js / React / TypeScript frontend, GraphQL middleware, OpenSearch data) — the direct foundation for the AI-powered self-serve dashboard above, re-using the same data contract.

SIDE PROJECTS

Clinical-Trials Agent — Grounded conversational research over ClinicalTrials.gov

Live: clinical-trials-agent-xi.vercel.app · Source: github.com/MBailey114/clinical-trials-agent

A dual-stack RAG agent (FastAPI + LangGraph backend, Next.js 16 frontend) that answers plain-language questions about clinical trials with per-claim citations validated server-side — so it can't cite a trial it never retrieved.

- **Grounded agent (4 tools)** over a deterministic 5-stage retrieval funnel: hybrid ranking blends a lexical score with optional OpenAI embeddings (cosine similarity), degrading cleanly to pure lexical under a latency budget, and token-bounded study cards keep the grounding set within budget.
- **Visualisations from the retrieved set** — the agent renders answers as grounded charts (Gantt timeline, comparison matrix, facet bars) in custom SVG, plus highlight-to-ask on any passage and a live "thought for..." reasoning trace.
- **Production concerns** — conversation persistence (Postgres) with Redis-backed resumable streaming (refresh mid-answer replays then live-tails the same run), per-IP rate limiting, and a three-pane research workspace with server-side snapshot pagination over the full matched set.

Stack: Python, FastAPI, LangGraph, Next.js 16, React 19, Vercel AI SDK v6, PostgreSQL, Redis, OpenAI (gpt-5.5 + text-embedding-3-small). Deployed on Vercel (frontend) + Fly.io (backend) with Neon Postgres and Upstash Redis; planned with an OpenSpec spec-driven workflow.

Ask F1 — Conversational F1 dashboard + race-replay viewer

Live: f1-dashboard-bay.vercel.app · Source: github.com/MBailey114/f1-dashboard

Public version of the Sky self-serve pattern, rebuilt on the Vercel AI SDK with no agent framework. A chat agent with 6 typed Zod tools mutates a 12-column dashboard grid in real time — one prompt streams a 4–6 widget themed dashboard — fronted by a custom agent step timeline (collapses to a pipeline pill, expands to per-tool input/output inspectors). A second surface (/tracks) replays pre-baked 2025 lap telemetry at 60Hz with Catmull-Rom interpolated car motion, multi-driver speed/throttle/brake charts, and a live AI race-commentary HUD. Cmd+K palette and LZ-compressed shareable URLs throughout; pivot logic is pure TypeScript — the model sees only field names and three sample rows, never the records.

Stack: Next.js 16, React 19, Vercel AI SDK v6, TypeScript strict, Recharts 3, Tailwind 4, Zustand 5, Zod 4, d3-geo, cmdk; three OpenAI tiers (gpt-5.5 / gpt-5.4 / gpt-5.4-nano), Edge runtime on every route, deployed on Vercel.

EDUCATION

Software Developer (Level 4) Apprenticeship — Multiverse / Sky UK

Sep 2022 – Apr 2024 · Distinction

Completed the Software Developer Level 4 apprenticeship standard (Institute for Apprenticeships and Technical Education) while working full-time in Sky's engineering organisation. Training provider: Multiverse. End-point assessment awarded Distinction on 24 April 2024. Work-based project: Sky's internal monitoring dashboards (Next.js / React, GraphQL, OpenSearch).